

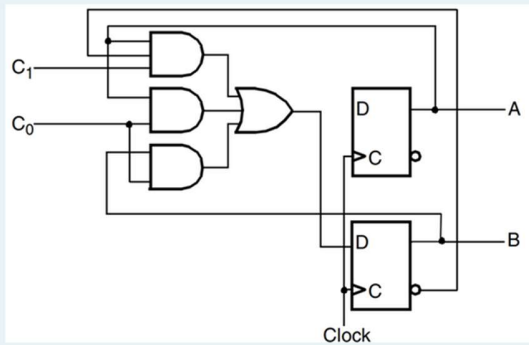
Question 1

Correct

Mark 17.50 out of 17.50

Flag question

Referring to the register cell transfer system below, determine the transfer operation which will happen if the control combination is $C_1C_0 = 01$ and $C_1C_0 = 10$.



- ☒ a. $C_1C_0 = 01: B \leftarrow (A \text{ or } B)$, and $C_1C_0 = 10: B \leftarrow (A \text{ and } \overline{B})$
- ☐ b. $C_1C_0 = 01: B \leftarrow (A \text{ and } B)$, and $C_1C_0 = 10: B \leftarrow (A \text{ or } \overline{B})$
- ☐ c. $C_1C_0 = 01: B \leftarrow (A \text{ and } \overline{B})$, and $C_1C_0 = 10: B \leftarrow (A \text{ or } B)$
- ☐ d. $C_1C_0 = 01: B \leftarrow (A \text{ or } \overline{B})$, and $C_1C_0 = 10: B \leftarrow (A \text{ and } B)$

Question 2

Correct

Mark 15.00 out of 15.00

Flag question

Compute the difference: $011011 - 001101$ using six-bit binary arithmetic by taking the 2s complement of the second operand and adding it?

- ☐ a. 010011
- ☒ b. 001110
- ☐ c. 011001

Your answer is correct.

$$011011 - 001101 = 011011 + (110010 + 1) = 011011 + 110011 = 001110$$

The correct answer is:

001110

Question 3

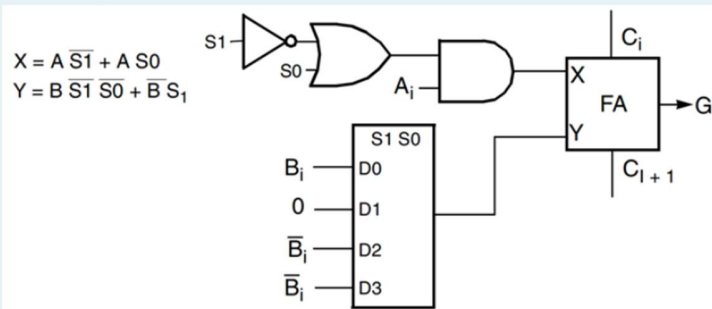
Correct

Mark 17.50 out of 17.50

Flag question

The logic diagram of a single bit stage of a 4-bit arithmetic circuit, with two selection inputs S_0 and S_1 , is shown below. Which arithmetic operation is performed for the given value of the selection inputs?

$S_1 S_0 = 10$, $C_{in} = 0$



- ☒ a. $G = \overline{B}$ (complement)
- ☐ b. $G = \overline{B} + 1$ (negate)
- ☐ c. $G = A + \overline{B} + 1$ (subtract)
- ☐ d. $G = A + B$ (add)

✓

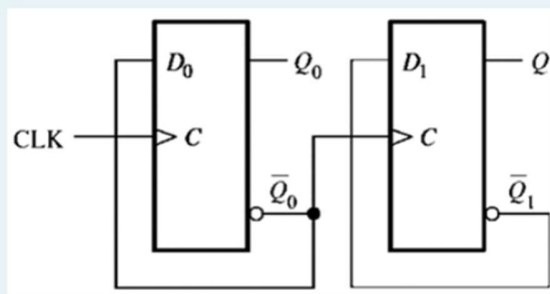
Question 4

Correct

Mark 17.50 out of 17.50

Flag question

For the positive edge-triggered ripple counter shown below, select the complete timing diagram during the 8-clock pulses for Q_0 and Q_1 .



- ☒ a.
- ☐ b.
- ☐ c.

✓

Question **5**

Correct

Mark 17.50 out of 17.50

🚩 Flag question

The content of a 4-bit register is initially 0101. The register is shifted eight times to the right, with the sequence 10110001 as the serial input. The leftmost bit of the sequence is applied first. What is the content of the register after each shift?

- ☐ a. 1001, 0101, 1110, 1101, 0110, 0011, 0001, 1000
- ☒ b. 1010, 0101, 1010, 1101, 0110, 0011, 0001, 1000
- ☐ c. 1110, 0101, 1010, 1011, 0110, 0011, 0001, 1000



Question **6**

Correct

Mark 15.00 out of 15.00

🚩 Flag question

A **1 Gb DRAM** uses **4-bit** data and has **equal-length** row and column addresses. How many **words** does the cell array contain? How many **address pins** does the DRAM have?

- ☒ a. The number of words: 2^{28}
The number of address pins: 14
- ☐ b. The number of words: 2^{30}
The number of address pins: 14
- ☐ c. The number of words: 2^{30}
The number of address pins: 15
- ☐ d. The number of words: 2^{18}
The number of address pins: 9

